

```

> restart;
with(student) :
with(linalg) :

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=
>

# =====
#      4
#      1
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# =====
#      4 . 4
#      : u_t t = u_x x, 0 < x < Pi, t > 0
#      : u(x, 0) = 0, u_t(x, 0) = 0, : u(0, t) = u(Pi, t) = 0
# =====

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> assume(n :: posint);

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pde := diff(u(x, t), t$2) = diff(u(x, t), x$2);
init := u(x, 0) = 0, D[2](u)(x, 0) = 0;
bound := u(0, t) = 0, u(Pi, t) = 0;



$$pde := \frac{\partial^2}{\partial t^2} u(x, t) = \frac{\partial^2}{\partial x^2} u(x, t)$$


$$init := u(x, 0) = 0, D_2(u)(x, 0) = 0$$


$$bound := u(0, t) = 0, u(\pi, t) = 0$$



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# : u(x, t) = X(x) * T(t)

subs(u(x, t) = X(x) * T(t), pde) :
expand(lhs(%) / (X(x) * T(t))) = expand(rhs(%) / (X(x) * T(t)));

# T''/T = X''/X = -lambda

lambda_eq_T := diff(T(t), t$2) + lambda * T(t) = 0;
lambda_eq_X := diff(X(x), x$2) + lambda * X(x) = 0;

```

(1)

$$\frac{\frac{d^2}{dt^2} T(t)}{T(t)} = \frac{\frac{d^2}{dx^2} X(x)}{X(x)}$$

$$\text{lambda_eq_T} := \frac{d^2}{dt^2} T(t) + \lambda T(t) = 0$$

$$\text{lambda_eq_X} := \frac{d^2}{dx^2} X(x) + \lambda X(x) = 0 \quad (2)$$

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(-)

solX := *dsolve*(*lambda_eq_X*, *X*(*x*));

X := *unapply*(*rhs*(*solX*), *x*);

e1 := *X*(0) = 0;

e2 := *X*(Pi) = 0;

sist := {*e1*, *e2*};

$$\text{solX} := X(x) = c_1 \sin(\sqrt{\lambda} x) + c_2 \cos(\sqrt{\lambda} x)$$

$$X := x \quad c_1 \cdot \sin(\sqrt{\lambda} \cdot x) + c_2 \cdot \cos(\sqrt{\lambda} \cdot x)$$

$$e1 := c_2 = 0$$

$$e2 := c_1 \sin(\sqrt{\lambda} \pi) + c_2 \cos(\sqrt{\lambda} \pi) = 0$$

$$\text{sist} := \{c_2 = 0, c_1 \sin(\sqrt{\lambda} \pi) + c_2 \cos(\sqrt{\lambda} \pi) = 0\} \quad (3)$$

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_EnvAllSolutions := *true*;

solve(*sin*(*sqrt*(*lambda*) * Pi) = 0, *lambda*);

subs(*_Z1* = *n*, %);

ev := *unapply*(%, *n*); # *lambda_n* = *n*²

$$\text{_EnvAllSolutions} := \text{true}$$

$$\text{_Z1} \sim^2$$

$$n \sim^2$$

$$\text{ev} := n \sim n \sim^2 \quad (4)$$

>

[0 , P i]

phi := (*n*, *x*) *sqrt*(2/Pi) * *sin*(*n***x*);

$$\phi := (n, x) \sqrt{\frac{2}{\pi}} \cdot \sin(x \cdot n) \quad (5)$$

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#

odeT := *diff*(*T*(*t*), *t*\$2) + *ev*(*n*) * *T*(*t*) = 0;

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solT := dsolve(odeT, T(t));
T_general := unapply(rhs(solT), t); # T(t) = _C1 cos(n t) + _C2 sin(n t)
```

$$odeT := \frac{d^2}{dt^2} T(t) + n^2 T(t) = 0$$

$$solT := T(t) = c_1 \sin(n \cdot t) + c_2 \cos(n \cdot t)$$

$$T_general := t \quad c_1 \cdot \sin(n \cdot t) + c_2 \cdot \cos(n \cdot t) \quad (6)$$

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u_series := Sum( ( A(n) * cos(n*t) + B(n) * sin(n*t) ) * phi(n, x), n = 1 ..infinity );
```

$$u_series := \sum_{n=1}^{\infty} \frac{(A(n) \cos(n \cdot t) + B(n) \sin(n \cdot t)) \sqrt{2} \sin(x \cdot n)}{\sqrt{\pi}} \quad (7)$$

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# u(x,0) = 0 => Sum( A(n)*phi(n,x) ) = 0 => A(n)=0
```

```
# u_t(x,0) = 0 => Sum( B(n)*n*phi(n,x) ) = 0 => B(n)=0
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A := n 0;
```

```
B := n 0;
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```
A := n 0
```

```
B := n 0
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(8)

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# ,
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u_solution := 0;
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```
u_solution := 0
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(9)

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printf(" ' ( 4 . 4 , 1 ) : \n " );
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printf("u(x,t)=0\n");
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' ( 4 . 4 , 1 ) :
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```
u(x,t) = 0
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> u := (x, t)  0;

                                     u := (x, t)  0                                (10)
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>
pde_residual := diff(u(x, t), t$2) - diff(u(x, t), x$2);

                                     pde_residual := 0                                (11)
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#
init_u := u(x, 0);
init_ut := eval(diff(u(x, t), t), t = 0);

                                     init_u := 0
                                     init_ut := 0                                (12)
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#
bound_left := u(0, t);
bound_right := u(Pi, t);

                                     bound_left := 0
                                     bound_right := 0                                (13)
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#
simplified_pde := simplify(pde_residual);

                                     simplified_pde := 0                                (14)
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```